

VCTriPop

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Overview

VCTriPop is a variation of VC-MAP-Elites that maintains three populations:

- Feasible (all constraints satisfied)
- Infeasible (static constraints not satisfied)
- Con Feasible (all static constraints satisfied; not all variable constraints satisfied)

To utilize memory, the infeasible population is permitted to grow until the feasible and constant constraint feasible populations fill their target memory allowance.

Populations

Feasible

Feasible individuals are stored in a QD array. Along the x-axis are bins determined by the diversity function, and each bin is sorted (on the y-axis) according to the fitness function.

Con Feasible

Like feasible individuals, con feasible individuals are stored in a QD array. In this case, each bin is sorted according to the modified fitness function:

$$F_c = [F * (1 - W_{vc})] + [F * W_{vc} * pct_{vc}]$$

where

F_c is the resulting fitness of the con feasible individual

F is the fitness given by the fitness function

W_{vc} is the weight assigned to the penalized fitness term $F * pct_{vc}$ (default .5)

$1 - W_{vc}$ is the weight assigned to the original fitness term F

pct_{vc} is the percentage of variable constraints that are satisfied

F_c is the resulting fitness, F is the fitness as determined by the fitness function, W_{vc}

Infeasible

Infeasible individuals are stored in a 1D array sorted according to the percentage of static constraints that are satisfied.

Memory Allocation

VCTriPop uses dynamic memory allocation. The feasible and con feasible populations are each allocated a portion of the maximum memory allowance, which is divided equally between the QD bins. Individuals are added to each bin until the allocated memory of the bin is exceeded, at which point the lowest fitness individual in the bin is removed. The infeasible population is allocated the remaining memory, but its population is allowed to grow as long as the feasible and con feasible populations do not use all of their allocated memory. At that point, the infeasible population is reduced according to a reverse roulette selection, in which lower fitness individuals are probabilistically more likely to be removed than higher fitness individuals.

Generation Cycle

Elitism

Each round of generation, the population is reshuffled if the variable constraints have changed. Then, the E most feasible “elites” of the infeasible population are kept, while the rest are culled according to the function:

$$E = \max(e * alloc, overflow),$$

where e is the elitism rate (default .7), $alloc$ is the memory allocation of the infeasible population, and $overflow$ is the amount by which the current infeasible population exceeds its memory allocation by extending into the feasible and con feasible allocation. Because the population grows by a limited amount each cycle, keeping the overflow allows us to use memory that otherwise would remain unfilled, while still using elitism to eliminate the least fit individuals.

Selection

Parents are selected according to a roulette selection procedure. First the population is chosen with probability according to the weighted population sizes. By default, the feasible population is given a weight of 2, and the con feasible population receives a weight of 1. If neither of those populations passes the probability check, then the infeasible population is chosen. In all cases, individuals are chosen from the population using roulette selection, in which higher fitness individuals are more likely to be chosen as parents.

Crossover and Mutation

New individuals are generated as in VC-MAP-Elites. A crossover operation is performed with probability given by the crossover rate, producing two children. Then, for each child (or each parent if there was no crossover), a mutation is performed with probability given by the mutation rate. Each resulting child is then placed in its population.